ALTERNATIVE INVESTMENT PERFORMANCE

Learning Outcome

- describe the performance appraisal of alternative investments

The unique features, form, and structure of alternative investments must be considered when evaluating the relative performance *between* alternative investments and when comparing their performance to that of more common asset classes over time. In particular, such features as staggered capital commitments over time, longer required investment horizons, reduced liquidity, and less efficient markets highlighted in prior lessons must be factored into the performance appraisal for alternative investments. Alternative investment returns are usually less normally distributed and therefore require different measures of risk and return than those used for more traditional asset classes.

Alternative Investment Performance Appraisal

Appraising the performance of alternative investments requires more scrutiny in certain areas than traditional asset classes do.

Comparability with Traditional Asset Classes

Public equity and debt securities share several characteristics that facilitate the comparison of their performance over a particular period. These standardized claims involve no further required capital commitments and provide identical claims to periodic cash flows, such as dividends in the case of shareholders or contractual bond coupons and principal for debtholders. Prices of publicly traded securities are often continuously quoted, with large peer groups of similar investments available and common indexes used to benchmark returns. Performance appraisal of publicly traded securities is thus straightforward to implement and evaluate.

In contrast, alternative investments are customized investments whose distinctive features complicate performance appraisal between investments and across asset classes. These features include

- the *timing* of cash inflows and outflows for specific investments,
- the use of borrowed funds,
- the *valuation* of individual portfolio positions over specific phases of the investment life cycle, and
- more complex fee structures and tax and accounting treatment.

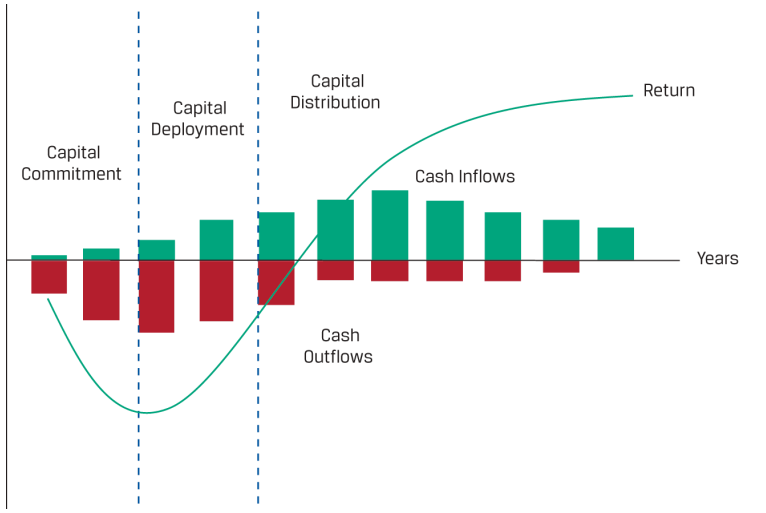
Performance Appraisal and Alternative Investment Features

When appraising alternative investments, four areas to focus on include the life cycle phase of the investment, the amount of borrowed funds used to maintain the market position, the valuation of the assets, and the fee structure of the fund.

Investment Life Cycle

Unlike public debt or equity securities purchased in the primary or secondary market, alternative investments usually involve a longer investment life cycle with distinct phases characterized by net cash outflows and inflows that complicate periodic return comparisons. Life cycle phases and timing vary across alternative investment types but generally fall into three distinct periods, as shown in Exhibit 1.

Exhibit 1:



- **Capital commitment:** Alternative managers identify and select appropriate investments with either an immediate or a delayed commitment of capital (known as a capital call) that may be in an early-stage company in the case of venture capital, a more mature firm for private equity, or one or more properties in the case of real estate. Returns are usually negative over this phase because fees and expenses are immediately incurred prior to capital deployment and assets may generate little or no income during this first phase.
- **Capital deployment:** Over this second phase, alternative managers deploy funds to engage in construction or make property improvements in the case of real estate or infrastructure, incur expenses in the turnaround phase of a mature company in the case of private equity, or initiate operations for a startup using venture capital. Cash outflows typically exceed inflows, with management fees further reducing returns.
- **Capital distribution:** When the turnaround strategy, startup phase, or property improvements are completed and if the investment is successful, the underlying assets appreciate in price and/or generate income in excess of costs, causing fund returns to accelerate. The fund may realize substantial capital gains from liquidating or exiting its investments, which may involve an initial public offering (IPO) for venture capital or the sale of properties in the case of real estate.

The so-called **J-curve effect** (because it resembles the letter J) shown in Exhibit 1 represents the initial negative return in the capital commitment phase followed by an acceleration of returns through the capital deployment phase. Returns often level off as capital is distributed to investors, investments are sold, and the fund is closed.

As a result of the cash inflows and outflows that occur over the investment life cycle, an internal rate of return is often used as an initial approach to calculate investment returns for these investments, which include private equity and real estate investments.

The performance assessment in both private equity and real estate investments depends far more on the timing and magnitude of cash flows in and out of the investments, and these are often hard to standardize and anticipate. Given the long time horizon, the application of different tax treatments can have a non-trivial impact on after-tax investment returns.

As a general rule, the best way to start evaluating such investments is with the IRR, taking into account the respective cash flows into an investment and the timing thereof, versus the magnitude and the timing of the cash flows returned by the investment (inclusive of tax benefits).

In an independent, fixed-life private equity fund, the decisions to raise money, take money in the form of capital calls, and distribute proceeds are all at the discretion of the private equity manager. Timing of cash flows is an important part of the investment decision

Although the determination of an IRR involves certain assumptions about a financing rate to use for outgoing cash flows (typically a weighted average cost of capital) and a reinvestment rate to use for incoming cash flows (which must be assumed and may or may not actually be earned), the IRR is the key metric used to assess longer-term alternative investments in the private equity and real estate worlds. IRR calculations will be covered in more depth in the Corporate Issuers topic area.

EXAMPLE 1

Peterburgh Capital, LLC

Peterburgh Capital, LLC, a private equity vehicle, is considering investment in various companies and is expecting the following cash flow pattern from these investments:

Year	Cash Inflow (USD)	Cash Outflow (USD)
0	—	2,500,000
1	—	4,000,000
2	—	1,700,000
3	500,000	1,000,000
4	1,000,000	—
5	3,000,000	—
6	12,500,000	—

IRR can be calculated by the standard formula $\sum_{t=0}^{n} \frac{CF_i}{(1+r)^i} = 0$, or using a financial calculator, r ≈ 14.596%.

As is true for any IRR calculation, assumptions regarding the opportunity cost of outgoing cash flows and the reinvestment rate for incoming cash flows will affect the results.

Because of this complexity, a shortcut methodology often used by both private equity and real estate managers involves simply citing a **multiple of invested capital (MOIC)**, or money multiple, on total invested capital (which is paid-in capital less management fees and fund expenses). Here, one simply measures the total value of all realized investments and residual asset values (assets that may still be awaiting their ultimate sale) relative to an initial total investment. MOIC is calculated as follows:

MOIC = (Realized value of investment + Unrealized value of investment)/Total am 1
of invested capital.

Although the MOIC ignores the timing of cash flows, it is easier to calculate, and it is intuitively easier to understand when someone says he received two or three times his initial investment. But how long it takes to realize this value does matter. A 2× return on one's initial investment would be great if the return were collected over two years but far less compelling if it took 15 years to realize.

In general, because private equity and real estate investments involve longer holding periods, there is less emphasis on evaluating them in terms of shorter-term portfolio correlation benefits. After a private equity fund has fully drawn in its monetary commitments, interim accounting values for a private equity partnership become less critical for a period of time because no incoming or outgoing cash flows may immediately hinge on such valuations. During this "middle period" in the life of a private equity fund, accounting values may not always be particularly reflective of the future potential realizations (and hence the expected returns) of the fund. It is not that the value of the investments is not actually rising and falling in the face of economic influences; rather, accounting conventions simply leave longer-lived investments marked at their initial cost

Knowledge Check: MOIC Calculation

Himitsu, a private equity firm, makes an initial investment of JPY3.8 billion into ZZZ company in Year 0. Eight years later, it sells its stake in ZZZ for JPY8.5 billion. Additional capital investments were made in Year 2 and in Year 3 for JPY1.2 billion and JPY200 million, respectively.

1. Calculate the MOIC.

Hide Solution

Solution:

MOIC = 8.5/(3.8 + 1.2 + 0.2) = 1.63×.

	Amount	Year
Invested Capital	(3,800,000)	0
	—	1
Additional Capital	(1,200,000)	2
Additional Capital	(200,000)	3
Liquidity Event	8,500,000	8
MOIC	**1.63×**	
IRR	**6.82%**	

Use of Borrowed Funds

Alternative investments may use borrowed funds to increase investment returns. This form of financial leverage has the effect of magnifying both gains and losses by allowing investors to take a market position that is larger than the capital committed. Consider a cash investment V_c with a periodic rate of return r. If we assume an investor is able to borrow at a periodic rate of r_b to increase the size of its investment by borrowed funds of V_b, we can calculate a simple *leveraged* rate of return r_L for the period as follows:

$$r_L = \text{Leveraged portfolio return/Cash position} = [r \times (V_c+V_b) - (V_b \times r_b)]/V_c. \quad 2$$

We can rearrange Equation 2 to show the relationship between the cash portfolio return, r, and the leveraged rate of return, r_L, as follows:

$$r_L = r + V_b/V_c(r - r_b). \quad 3$$

Hedge funds leverage their portfolios by using derivatives or borrowing capital from **prime brokers**, negotiating with them to establish margin requirements, interest, and fees in advance of trading. In a typical **margin financing** arrangement, the prime broker essentially lends the hedge fund the shares, bonds, or derivatives, and the hedge fund deposits cash or other collateral into a margin account with the prime broker based on certain fractions of the investment positions. The margin account represents the hedge fund's net equity in its positions. The minimum margin required depends on the riskiness of the investment portfolio and the creditworthiness of the hedge fund.

Leverage is a large part of the reason that some hedge funds either earn larger-than-normal returns or suffer significant losses. If the margin account or the hedge fund's equity in a position declines below a certain level, the lender initiates a margin call and requests that the hedge fund put up more collateral. An inability to meet margin calls can have the effect of magnifying or locking in losses because the hedge fund may have to liquidate (close out) the losing position. This liquidation can lead to further losses if the order size is sufficiently large to move the security's market price before the fund can sufficiently

value strategies. But with added leverage comes increased risk.

EXAMPLE 2

Leverage by Lupulus Opportunity Fund LLC

Lupulus Opportunity Fund LLC, a hedge fund that has USD100 million of capital, ordinarily uses leverage to invest in a variety of equity-linked notes.

Scenario 1: Suppose Lupulus's underlying positions return 8%. If it could add leverage of USD50 million to the portfolio at a funding cost of 4%, what would have been the leveraged return?

Using Equation 3, the leveraged return can be calculated as follows:

$V_c = 100$; $V_b = 50$.

$r_L = 0.08 + (50/100)(0.08 – 0.04) = 10\%$.

Scenario 2: Suppose Lupulus's underlying positions incur a loss of 2% instead of earning a gain. What would have been the leveraged return if Lupulus had borrowed USD50 million at 4%?

Again, using Equation 3, $V_c = 100$ and $V_b = 50$

$r_L = –0.02 + (50/100)(–0.02 – 0.04) = –5\%$.

In other words, Lupulus would have magnified its portfolio loss from –2% to –5% by adding leverage of USD50 million.

Scenario 3: If Lupulus's underlying positions make a gain of 6%, what is the breakeven borrowing rate at which Lupulus is indifferent to adding leverage to the portfolio?

From Equation 3, Lupulus would be indifferent about adding leverage when the magnification effect of leverage equals 0; that is,

when $r_L = r$, or $(V_b/V_c)(r – r_b) = 0$,

$r_b = r = 6\%$.

As the previous example shows, the leveraged rate of return will exceed that of an equivalent cash portfolio if the market return on assets exceeds the borrowing rate. This example assumes the investor has unlimited access to borrow freely on an unsecured basis. The access to borrowed funds or lack thereof is an important factor in evaluating the risk of leveraged investments.

Valuation

Alternative assets are often characterized by illiquidity, which makes performance appraisal over time challenging and periodic comparison with common asset classes difficult. Accounting rules dictate that investments must be recorded at their **fair value** for financial reporting purposes. The fair value of an investment is a market-based measure based on observable or derived assumptions to determine a price that market participants would use to exchange an asset or liability (often referred to as the *exit price* for a seller) in an orderly transaction at a specific time. Assumptions used to measure fair value follow a three-level hierarchy, as shown in Exhibit 2:

Exhibit 2:

Fair Value Hierarchy

Level 1	Quoted prices in active markets for identical asset/liability that may be accessed as of measurement date	Exchange-traded public equity securities (observed closing market price)
Level 2	Inputs other than quoted market prices in Level 1 that are directly or indirectly observable for an asset/liability	Over-the-counter interest rate derivatives (pricing model using quoted market prices)
Level 3	Unobservable inputs are used to measure fair value for asset/liability in which there is little, if any, market activity as of the measurement date	Private equity or real estate investments (cash flow projection models with reasonably available market participant assumptions)

While common asset classes often rely on Level 1 inputs, the interim valuation of private equity, real estate, and other less frequently traded assets based on Level 3 inputs is more challenging. Interim accounting values may be less critical for the partnership itself over a period where no incoming or outgoing cash flows are expected. The lack of new market information over time may thus anchor the value of these long-lived investments at or near initial cost, with adjustments to carrying value when impairments or realization events occur. The relatively stable accounting valuations of these strategies may give investors a false sense that they are less correlated with and less volatile than other investments, although a more realistic picture may emerge if managers are forced to liquidate a portfolio prematurely.

For Level 3 asset pricing, regardless of the model used by a manager in such circumstances, that model should be independently tested, benchmarked, and calibrated to industry-accepted standards to ensure a consistency of approach. Because of the potential for conflicts of interest when applying estimates of value, hedge funds must develop procedures for in-house valuation, communicate these procedures to clients, and adhere to them consistently.

Notwithstanding best practice, the very nature of assets that can be valued only on a “mark-to-model” basis can and should be a focus for the alternative asset investor. A model may reflect an imperfect theoretical valuation and not a true liquidation value. The illiquid nature of these assets means that estimates, rather than observable transaction prices, may well have factored into any valuation. As a result, returns may be smoothed or overstated and the volatility of returns, understated. As a generalized statement, any investment vehicle that is heavily involved with Level 3 priced assets deserves increased scrutiny and due diligence.

Fees

Alternative investment fees also vary from those for common asset classes, which typically involve a flat management fee. Alternative investments often levy additional performance fees based on a percentage of periodic fund returns. Performance appraisal for these investments can be difficult to generalize, because results may vary significantly based on *which* investor has invested *when* in a particular vehicle.

For example, an investor may face significantly lower incentive fees if she invests *more* capital in a fund at an *earlier* phase or is willing to accept *greater* restrictions on redemptions. Also, an investor entering an alternative fund following a sharp drop in value may incur performance fees if the fund rises, while an earlier investor who experienced the sharp decline in value from its peak may be exempt from such fees for the same period. These and other details of these complex alternative investment fee arrangements and their effect on investment return are the subject of the following lesson.

Question Set

1. Which of the following is *not* a factor that makes comparison of performance between alternative investments and public securities difficult?

A. Alternative investments charge higher fees.

C. The fair value of portfolio positions in alternative investments may not be readily available.

Hide Solution

Solution:

A is correct. Although an alternative investment may charge a higher fee, it is not the absolute fee level but the complexity of the fee arrangement that makes alternative investment appraisal unique compared to other common asset classes.

Alternative investments often involve the use the of explicit leverage, which has the effect of magnified gains and losses. Alternative assets are often characterized by illiquidity with unobservable market prices, making performance appraisal over time and periodic comparison with common asset classes challenging.

2. Describe the J-curve effect in alternative investments.

Hide Solution

Solution:

The J-curve effect in alternative investments describes the initial negative return in the capital commitment phase followed by an acceleration of returns through the capital deployment phase. Returns often level off as capital is distributed to investors, investments are sold, and the fund is closed.

3. A private equity closes a fund with a capital commitment of €750 million. It has a capital call of €500 million initially and another €250 million at the end of Year 1. The management fee is 2% per annum. At the end of Year 5, a total of €1.0 billion is distributed to its investors, and the fund is left with €500 million in asset value. The multiple of invested capital (MOIC) after five years is *closest* to:

A. 1.3×.
B. 2.0×.
C. 2.2×.

Hide Solution

Solution:

C is correct. Using Equation 1, MOIC = (Realized value of investment + Unrealized value of investment)/Total amount of invested capital, where invested capital equals total paid-in capital less management fees and fund expenses. MOIC is different from the IRR measure because it ignores the timing of cash flows.

Total paid-in capital = 500 + 250 = 750.

Total management fee for 5 years = 750 × 0.02 × 5 = 75.

Total invested capital = 750 – 75 = 675.

MOIC = (1,000 + 500)/675 ≈ 2.2×.

4. If the market return of underlying portfolio positions is expected to be 10% and the fund manager adds 100% explicit leverage to its capital at a borrowing rate of 3%, what is the expected leveraged return?

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Solution:

Using Equation 3,

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5. Match the fair value classification of assets to their description for financial reporting purpose:

Fair Value Classification Description

1. Level 1 assets	(a) Inputs other than quoted market prices are directly or indirectly observable for valuation on measurement date
2. Level 2 assets	(b) Unobservable inputs are used to evaluate fair value for the assets
3. Level 3 assets	(c) Quoted prices in active markets for identical asset available on measurement date

Hide Solution

Solution:

1. Level 1 assets: (c) Quoted prices in active markets for identical asset available on measurement date

2. Level 2 assets: (a) Inputs other than quoted market prices are directly or indirectly observable for valuation on measurement date

3. Level 3 assets: (b) Unobservable inputs are used to evaluate fair value for the assets

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Discussion

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AC Question set 3: is this correct? MOIC is calculated based on ex-fees in the answer C. Shouldn't MOIC be calculated inclusive of fees?

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MD For the MOIC Knowledge Check Question isn't the IRR 6.82. I used the CF function on BA II Plus. CF 0 -3.8, CF 2 -1.2, CF 3 -0.2 and CF 8 8.5 gives IRR of 6.82

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